

AICTE | IBM SkillsBuild Data Analytics with AI Internship 2026

Organization: BharatCares in association with AICTE

Project Report: Customer Churn Prediction & Executive Retention Dashboard

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Domain: Customer & Marketing Analytics

1. Executive Summary & Problem Definition

- **Business Challenge:** In competitive service industries like telecommunications, acquiring new customers costs significantly more than retaining existing ones. Unmonitored attrition leads directly to substantial lost recurring revenue.
- **Project Objective:** Build an end-to-end data analytics and AI solution following the 7-stage data lifecycle. The objective is to identify key drivers of attrition, proactively predict churn probability using a Random Forest machine learning model, and deliver an interactive executive dashboard with prescriptive retention strategies for decision-makers.

System Architecture & 7-Stage End-to-End Pipeline

Stage	Lifecycle Phase	Key Deliverables & Methods
Stage 1	Problem Definition	Define business challenge, churn KPI, and revenue retention scope

Stage	Lifecycle Phase	Key Deliverables & Methods
Stage 2	Data Preprocessing	IBM Telco dataset loading, whitespace handling, median imputation, one-hot encoding
Stage 3	Exploratory Data Analysis	Identify contract type risks, tenure vulnerability, and billing tier correlations
Stage 4	AI / Predictive Modeling	Train Random Forest Classifier (100 trees, depth 6) with balanced precision/recall
Stage 5	Risk Scoring Engine	Generate calibrated churn probabilities (0.0 to 1.0) per customer
Stage 6	Executive BI Dashboard	Streamlit web dashboard tracking Total Accounts, Churn %, and At-Risk MRR (\$)
Stage 7	Strategic Recommendations	Actionable retention plays: annual contract incentives, 90-day onboarding workflows

2. Data Sourcing & Preprocessing (Stages 1–2)

- **Dataset:** IBM Telco Customer Churn open tabular dataset comprising 7,043 customer records across 21 demographic, service, and account attributes.
- **Data Cleaning Steps:**
 - a. Handled missing and whitespace values in **TotalCharges** by converting to numeric format and performing median imputation.
 - b. Encoded the target variable **Churn** into a binary format (1 for Churn, 0 for Retained).
 - c. Formatted categorical predictors (**Contract**, **InternetService**, **PaymentMethod**, **TechSupport**) via one-hot encoding for machine learning readiness.
 - d. Segmented tenure and monthly charge metrics to evaluate lifetime value and vulnerability windows.

3. Exploratory Data Analysis (EDA) Insights (Stage 3)

- **Contract Type Impact:** Customers on month-to-month contracts exhibit an attrition rate exceeding 42%, whereas one-year and two-year contract holders churn at ~11% and ~3% respectively.
- **Tenure Risk Window:** The highest churn density occurs within the first 1 to 6 months of customer tenure, highlighting a critical onboarding gap.
- **Service Configuration:** Accounts utilizing Fiber Optic internet without supplementary tech support or device protection show heightened churn sensitivity tied to higher monthly billing tiers.

4. AI & Predictive Modeling (Stages 4–5)

- **Model Architecture:** Supervised Random Forest Classifier (100 estimators, max depth of 6 to prevent overfitting and ensure interpretability).
- **Evaluation:** Achieved ~80% test accuracy with balanced precision-recall on at-risk customer detection.
- **Risk Scoring Engine:** Model outputs continuous probability scores (0.0 to 1.0) categorized into operational risk tiers:

Risk Tier	Churn Probability	Primary Driver Profile	Prescriptive Strategic Action
 Low Risk	< 0.35	Long tenure, two-year contracts, bundled tech support	Value protection, loyalty perks, cross-sell/upsell
 Medium Risk	0.35 – 0.65	Intermediate tenure (6–18 mos), electronic check users	Automated usage check-ins, satisfaction surveys
 High Risk	> 0.65	Month-to-month contracts, fiber optic without tech support	Immediate proactive intervention, 10% annual migration discount

5. Executive Dashboard & Strategic Recommendations (Stages 6–7)

- **Dashboard Interface:** Built an interactive web dashboard in Streamlit displaying high-level executive KPIs: Total Monitored Customers, Aggregate Churn Rate, and Total Monthly Recurring Revenue (MRR) at High Risk.
- **Actionable Strategic Recommendations:**
 - Contract Migration Campaigns:** Offer targeted incentives (e.g., 10% annual loyalty discounts) to transition high-risk month-to-month subscribers into annual contracts.
 - Enhanced 90-Day Onboarding:** Deploy automated proactive check-ins and onboarding support touchpoints during the first 3 months of tenure.
 - Targeted Retention Allocation:** Focus marketing and concession budgets strictly on predicted "High Risk" accounts with monthly charges > \$70 to protect revenue while avoiding unnecessary discounts for loyal customers.

6. Technical Appendix

